

题目

已知 $D(X) = 4$, $D(Y) = 1$, $\rho_{XY} = 0.6$, 则 $D(3X - 2Y) =$ _____

步骤

对于两个随机变量的线性组合, 方差计算公式为:

$$D(aX + bY) = a^2 D(X) + b^2 D(Y) + 2ab \cdot \text{Cov}(X, Y)$$

其中协方差 $\text{Cov}(X, Y)$ 通过相关系数求得:

$$\text{Cov}(X, Y) = \rho_{XY} \cdot \sqrt{D(X)} \cdot \sqrt{D(Y)}$$

题解

代入题目: $D(X) = 4$, $D(Y) = 1$, $\rho_{XY} = 0.6$

先求协方差:

$$\text{Cov}(X, Y) = 0.6 \times \sqrt{4} \times \sqrt{1} = 0.6 \times 2 \times 1 = 1.2$$

代入方差公式 ($a = 3$, $b = -2$):

$$D(3X - 2Y) = 3^2 \times 4 + (-2)^2 \times 1 + 2 \times 3 \times (-2) \times 1.2$$

$$= 9 \times 4 + 4 \times 1 - 14.4$$

$$= 36 + 4 - 14.4 = 25.6$$

所以 $D(3X - 2Y) = \mathbf{25.6}$